

MUHAMMAD ZAYAB ANSARI

AI Engineer

+92 3333450915 • zayabansari@outlook.com • Nazimabad 2, Karachi, Pakistan
github.com/Zay7ab • linkedin.com/in/muhammad-zayab-ansari-33947a121/

PROFESSIONAL SUMMARY

Software Engineering graduate specialising in AI engineering and scalable machine learning systems. Experienced architecting RAG pipelines, agentic AI workflows, and cloud-native model deployment using FastAPI, LangChain, Docker, and LLMs. Delivered end-to-end AI systems spanning NLP, Computer Vision, and Generative AI — from data pipeline design and model fine-tuning to containerised production deployment with model monitoring. Proficient in MLOps best practices including CI/CD via GitHub Actions, inference optimisation, and experiment tracking. Seeking a junior AI Engineer role to build impactful, scalable AI products.

EXPERIENCE

AI Engineering Intern | SMIT Bootcamp 5.0

Jan 2025 – Present

- Engineered and deployed 10+ AI models across NLP, Computer Vision, and Generative AI domains; containerised via Docker to Hugging Face Spaces and cloud-native environments, reducing deployment time by ~60% using CI/CD pipelines.
- Architected end-to-end ML data pipelines covering preprocessing, feature engineering, model training, hyperparameter tuning, and scalable model deployment; versioned all artefacts with structured experiment tracking logs.
- Integrated LLM-based agentic AI solutions with FastAPI backends and Streamlit frontends; enforced REST API versioning and automated testing via GitHub Actions, cutting manual QA effort by 40%.
- Applied inference optimisation techniques (ONNX export, quantisation) reducing average model inference latency by 35% across deployed Computer Vision models.

Freelance AI Engineer | Fiverr (Self-Employed)

Mar 2024 – Present

- Delivered multiple end-to-end ML, NLP/RAG, and Computer Vision solutions for international clients across healthcare, legal, and e-commerce verticals; maintained 5-star rating across all engagements.
- Architected and deployed custom AI solutions including RAG chatbots, image classifiers, and Generative AI integrations on client-managed AWS/GCP-compatible cloud infrastructure using Docker and scalable microservices.
- Reduced average client project delivery time by 30% through reusable ML pipeline templates covering data ingestion, model fine-tuning, evaluation, and containerised cloud deployment.

EDUCATION

B.E. Software Engineering | SSUET, Karachi

2022 – 2026

- Final Year Project: Pakistan LegalAI Platform — production-ready RAG-based legal AI system achieving 95%+ retrieval accuracy across 12+ laws; deployed via Docker + Nginx with full CI/CD pipeline.
- Relevant coursework: Big Data Analytics (Hadoop, Spark, MapReduce), Machine Learning Fundamentals, Database Systems, Computer Networks, OOP, System Design.

Data Science & AI Bootcamp | SMIT — Saylani Mass IT Training

2025 – 2026

- Completed intensive training in Deep Learning (CNNs, Transformers, Transfer Learning), LLMs, RAG pipelines, agentic AI workflows, model fine-tuning, and MLOps fundamentals including model monitoring and experiment tracking.

PROJECTS

Sentrix — Enterprise AI Security Platform (In Progress)

2025 – 2026

Python, FastAPI, SQLAlchemy, XGBoost, scikit-learn, LSTM, Docker, WebSocket, React

- Architecting a cloud-native Security Operations Center (SOC) platform with a 7-model AI ensemble achieving real-time threat detection across network traffic; 300+ automated tests ensure >98% pipeline reliability.
- Engineered ML data pipeline combining XGBoost, Isolation Forest, LSTM, and behavioural analytics for ransomware, C2, and brute-force detection; mapped all detections to MITRE ATT&CK framework with automated SOAR playbooks.
- Implemented scalable microservices backend with JWT authentication, RBAC, async PostgreSQL, and WebSocket-based real-time alerting; supports 10 automated incident response playbooks reducing MTTR by design.
- Multilingual marketing website supporting 10 languages with RTL layout, dark/light mode, and fully responsive design.

Python, FastAPI, React 18, LangChain, FAISS, LLaMA 3-70B, Groq API, PostgreSQL, Docker, Nginx, Tailwind CSS

- Engineered a production-ready RAG pipeline ingesting 12+ Pakistani laws into a FAISS vector index; achieved 95%+ retrieval relevance with LLaMA 3-70B via Groq API and average response latency under 3 seconds.
- Delivered 25+ JWT-authenticated REST API endpoints, Urdu/English bilingual chatbot, court case search across 21 real judgments (PLD/SCMR), legal PDF generation, and a Know Your Rights module.
- Implemented full MLOps lifecycle: Docker Compose containerisation, CI/CD via GitHub Actions, structured API versioning, and environment-based config management for scalable cloud deployment.

ClinIQ — Clinical AI Platform

2024

Python, FastAPI, Streamlit, Random Forest, CNN/ONNX, LLaMA 3.3-70B, Docker, Hugging Face Spaces

- Built a multimodal clinical AI platform integrating Random Forest, CNN/ONNX (inference-optimised), and LLaMA 3.3-70B across 3 independent microservices; reduced diagnostic query response time to under 2 seconds per request.
- Deployed each service as an independently containerised unit on Hugging Face Spaces with model monitoring hooks tracking inference latency and prediction confidence in real time.

Real-Time Brain Tumor Detection System

2024

Python, YOLOv11 (Ultralytics), ONNX, Streamlit, Hugging Face Spaces

- Trained YOLOv11 model via transfer learning and data augmentation achieving 93.5% mAP at 30 FPS on medical MRI streams; exported to ONNX for inference optimisation and cross-platform scalable model deployment.
- Shipped interactive Streamlit interface for real-time annotation and upload; deployed publicly on Hugging Face Spaces with zero-downtime containerised hosting.

Flight Delay Prediction System

2024

Python, XGBoost, Scikit-Learn, SMOTE, MLflow, Pandas, NumPy

- Designed end-to-end ML data pipeline for flight delay prediction; achieved 89% precision/recall with XGBoost via temporal and categorical feature engineering; tracked all experiments using MLflow for reproducibility.
- Resolved class imbalance via SMOTE; benchmarked 5 classification algorithms with cross-validation — XGBoost outperformed all baselines by 12% on F1-score.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, SQL, HTML/CSS
AI / ML	Scikit-Learn, TensorFlow, PyTorch, XGBoost, LightGBM, Random Forest, CNNs, Transfer Learning (VGG16, MobileNetV3), SMOTE, Transformers, model fine-tuning, inference optimisation
Generative AI	LLMs, Prompt Engineering, RLHF (concepts), OpenAI API, Groq API, Hugging Face, model fine-tuning, RAG pipelines, local LLM inference (Ollama), embeddings
Agentic AI	LangChain, LangGraph, agentic AI workflows, FAISS, ChromaDB, Sentence Transformers, vector databases, semantic search
Computer Vision	YOLOv11, ONNX, OpenCV, image preprocessing, data augmentation, feature extraction, MRI analysis, real-time inference
Voice & Speech	Whisper (STT), Coqui TTS, voice cloning, audio preprocessing
Backend & APIs	FastAPI, REST API design & versioning, SQLAlchemy, PostgreSQL, SQLite, Redis, JWT auth, Docker, Docker Compose, Nginx, WebSocket, scalable microservices, cloud-native deployment
MLOps	CI/CD (GitHub Actions), model deployment, model monitoring, experiment tracking (MLflow), model versioning, data pipeline design, containerisation, Hugging Face Spaces, AWS/GCP
Frontend	Next.js, React 18, TypeScript, Tailwind CSS, Streamlit, Gradio
Data & Analytics	EDA, feature engineering, A/B testing, predictive modelling, data visualisation (Matplotlib, Seaborn), Pandas, NumPy
Tools	Git/GitHub, Jupyter Notebook, VS Code, Linux/Bash, Redis, Postman

CERTIFICATIONS & ACHIEVEMENTS

Cisco Networking Academy

- Python Essentials 1 — Dec 2025
- Python Essentials 2 — 2026

Kaggle

- Computer Vision • Intro to Deep Learning • Intro to Machine Learning • Data Cleaning • Data Visualization • Python

Saylani Mass Training Program

- Coding Night 2026 — Hackathon Participant